

Divyanshu Chauhan

+91-9548022166 | divyanshuchauhan1200@gmail.com | Noida, Uttar Pradesh | LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Full Stack Software Engineer with 2+ years of professional experience building end-to-end web applications and developer-facing platforms using **JavaScript/TypeScript, React, Node.js, Python**, and **REST APIs**, deployed on **AWS** with **Docker** and **CI/CD pipelines**. Proven track record shipping production systems at scale — including a **White-Label RAG SaaS platform** and a **defence-sector Smart Inventory Tracking Solution (HAL Korwa)** — with measurable impact: 40% improvement in retrieval accuracy, 50% reduction in processing latency, and 80% faster deployments. Strong advocate for developer experience, accessible UI design, and operational excellence across the full stack.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, SQL, Bash, Node.js

Frontend: React, HTML5, CSS3, Tailwind CSS, Responsive & Accessible Web Design

Backend & APIs: Node.js, FastAPI, Flask, REST API Design, JWT Auth, WebSockets, Microservices

Databases & Storage: PostgreSQL, MySQL, MongoDB, Redis, pgvector, Celery

Cloud & DevOps: AWS (EC2, S3), Docker, Jenkins, GitLab CI, CI/CD Pipelines, Serverless Architecture

Observability & Tools: Grafana, ELK Stack, Postman, Git/GitHub, Agile/Scrum, Jira

AI/ML & GenAI: RAG Systems, LLM Integration, LangChain, LangGraph, Embeddings, Vector Search, Ollama

Design Systems: Component Library ecosystems (Tailwind CSS; familiar with React Spectrum patterns)

EXPERIENCE

Full Stack Developer (Python + AI) | *Infosware Private Limited*

Noida, India

- Architected and shipped a production-grade **White-Label RAG SaaS platform** (FastAPI, React frontend, pgvector, Redis, Celery, Ollama) — full pipeline from document ingestion → embeddings → hybrid retrieval → LLM response, serving as an end-to-end full stack system.
- Designed and exposed **REST APIs** consumed by React dashboards and third-party WhatsApp chatbot integrations; implemented JWT-based multi-tenant authentication and role-based access control.
- Increased retrieval accuracy by **40%** by designing multi-tier RAG pipelines (Naive / Simple / Advanced) with intelligent routing, maintaining <2s P95 latency in production.
- Eliminated **60% false retrievals** by diagnosing BM25 index staleness and implementing **event-driven** automatic re-indexing with Celery workers — a microservices pattern aligned with Adobe's I/O Events architecture.
- Reduced document processing time by **50%** through distributed async architecture (Redis + Celery), enabling horizontal scaling across workers.
- Cut AI inference cost by **25%** using prompt compression, response caching, and query deduplication driven by Prometheus performance analysis.
- Launched **multilingual support for 10+ Indian languages** (Indic LID + IndicTrans2), enabling large-scale WhatsApp chatbot deployments used by SMB clients.
- Backend Developer** — **Smart Inventory Tracking Solution (SITS), HAL Korwa Avionics Division** (defence-sector programme via Thinkbotic, Infosware's parent company): designed **project- and batch-aware master data models** (projects, vendor-invoice-derived batches, five-level storage hierarchy) and built **zone-based Transfer ID send/receive reconciliation services** matching RFID/barcode reads across every inter-department doorway (match / short-receipt / excess-receipt outcomes), replacing manual re-counts at each department handoff.

QA Software Engineer | *Tekshapers (Magic EdTech)*

Noida, India

- Designed a **Python NLP pipeline** automating ALT-text generation for EPUB accessibility content, reducing manual validation from 3 days to 2 hours (**95% reduction**); demonstrates commitment to **accessible web design** aligned with WCAG standards.
- Diagnosed and resolved a **production race condition** in concurrent database record lookups causing QTI assessment launch failures; documented fix, preventing recurrence across 3 product teams.
- Built an **automated regression testing framework** for the SPOUT standards-mapping engine, improving test coverage efficiency by **40%** and reducing production defect leakage by **30%**.
- Engineered pre-release validation workflows ensuring **99.9% release reliability**; achieved zero critical post-release defects across consecutive cycles, earning formal client recognition.

DevOps Intern | *AR Electricals*

Noida, India

- Designed and implemented **Jenkins-based CI/CD pipelines** from scratch for multi-environment deployments, reduc-

ing release cycle time by **70%** (45 min → 13 min).

- Developed automated build, test, and **containerized deployment workflows** using Bash and Docker, eliminating 50% of manual steps and standardizing environment parity across dev, staging, and production.

PROJECTS

RAG AI Chatbot SaaS Platform — Developer-Facing API Platform

2026

Tech: Python, FastAPI, React, Node.js, pgvector, Redis, Celery, Ollama, LangChain, AWS EC2/S3, Docker

- **Challenge:** Indian SMBs needed a multilingual AI chatbot platform without per-query GPT-4 costs; built a **self-hosted, white-label SaaS** with local LLMs eliminating per-token API costs entirely.
- Exposed a **unified REST API** behind three RAG pipelines (Naive for speed / Simple for accuracy / Advanced with re-ranking for quality) — clients get SLA-based routing via a developer-friendly API surface, similar to Adobe's App Builder extension model.
- Solved the **chunk boundary problem** (semantic search splitting context mid-sentence) using parent-child chunking: small child chunks for retrieval precision, large parent chunks for LLM context quality.
- **Outcome:** 35% improvement in response relevance, 50% reduction in processing time, -40% token cost vs GPT-4 baseline.

Smart Inventory Tracking Solution (SITS) — HAL Korwa Avionics Division

2026

Tech: Python, Node.js, FastAPI, PostgreSQL, RFID/IoT Integration, REST APIs

- **Challenge:** HAL Korwa's defence-grade inventory (2 transit + 3 holding stores) required project- and batch-level traceability across departments without duplicate manual counting.
- Designed **backend services** for the master data model — projects, vendor-invoice-derived batches, and five-level storage hierarchy (Room → Shelf → Row → Position → Bin).
- Built **zone-based Transfer ID reconciliation logic** anchoring send-side and receive-side RFID/barcode reads at every inter-department doorway, producing deterministic *match* / *short-receipt* / *excess-receipt* outcomes.
- **System design decision:** split FIFO issue logic by project-tracked vs. shared-item flags; cross-project draws require explicit Store-in-Charge approval, balancing per-project accounting accuracy against shared-pool efficiency.
- **Outcome:** replaced manual re-counts at each department handoff with automated reconciliation — delivered via Thinkbotic (Infosware's parent company) for a live defence-sector deployment.

Indian Army Logistics System

2026

Tech: Python, Flask, React, RFID/NFC, MIFARE 1K, PostgreSQL, WebSocket

- Designed an enterprise-grade military logistics system that manages the entire inventory lifecycle (delivery of the warehouse → field), 4 user roles with RBAC, and real-time multi-client synchronization via **WebSocket**.
- Built **RESTful Flask API** (15+ endpoints) and a **normalized PostgreSQL database** with 8 tables and automated triggers for real-time stock calculations, reducing manual data entry errors by **100%**.
- Designed smart allocation algorithm auto-locating items across warehouse hierarchy (Warehouse → Sheds → Containers → Boxes) based on availability and deployment priority.

CI/CD Deployment Pipeline

2025

Tech: Flask, Express (Node.js), AWS EC2/S3, Docker, Jenkins

- Automated a 30-minute manual deployment process in a build pipeline → test → dockerize → with a **blue-green deployment strategy**, achieving zero-downtime releases.
- **Result: 80% reduction** in deployment time (30 min → 5 min), zero failed production deployments after implementation.

EDUCATION

Master of Computer Applications – Gurukul Kangri Vishwavidyalaya

CGPA: 8.62 | Haridwar

Relevant Coursework: Advanced Algorithms, Software Engineering, Computer Networks, Database Management, AI/ML Fundamentals

Bachelor of Science (Computer Science) – Gurukul Kangri Vishwavidyalaya

CGPA: 8.46 | Haridwar

Relevant Coursework: Data Structures, Cloud Computing, DBMS, Linux, Software Development

CERTIFICATIONS

- IBM RAG and Agentic AI Professional Certificate – IBM May 2026
- Agentic AI with LangChain and LangGraph – IBM May 2026
- Develop Generative AI Applications: Get Started – IBM May 2026
- Oracle Cloud Infrastructure 2025 Certified DevOps Professional
- Microsoft Introduction to DevOps